

3) While :**Problem :**

- Enter a strictly positive integer x then find out how many times it is divisible by 2.

Example : 8 is divisible 3 times by 2 ($2*2*2$).

Solution :

Repeat

Write ("Give a strictly positive integer : "); Read (x) ;

Until (x>0) ;

cpt ← 0;

While (x mod 2 = 0) do

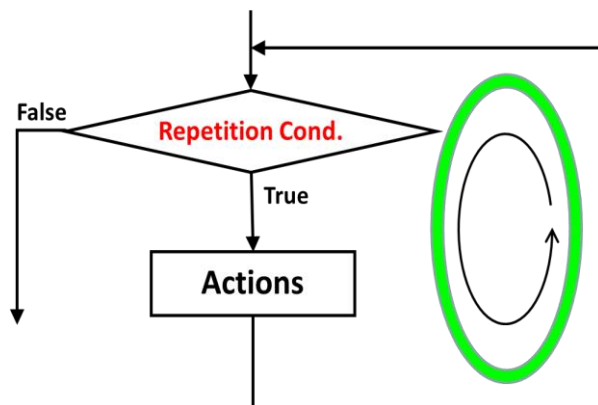
 cpt ← cpt+1; x ← x div 2;

EndWhile;

Print("cpt=", cpt);

Syntax :

While (repetition condition) do
Actions;
EndWhile;



Where :

repetition condition : It is a Boolean expression, which designates the case of repetition of actions.

Remarks :

- ❖ The number of repetitions is unknown, because it depends on a condition, we can repeat the loop 1 time, 5 times, 10,000,000 times, for a week, a year,... We only leave the loop when the condition is false.
- ❖ The actions of the loop may not be executed (the actions executed out at least zero times).

General Conclusion